7.8.3 Connections between Analog and Digital Carrier Modulations

- Because ASK uses on-off signaling (nonnegative amplitude), it is essentially an AM signal with modulation index μ = 1.
- FSK is actually an FM signal with only limited number of selections as instantaneous frequencies.
- For PSK, the modulated signal can be written as

$$\varphi_{\mathrm{PSK}}(t) = A\cos(\omega_c t + \theta_k) \qquad kT_b \le t < kT_b + T_b$$

In particular, since binary PSK uses θ = 0, π, we can write binary PSK signal as $\pm A cos(\omega_c t)$

So, PSK is effectively a digital manifestation of the DSB-SC amplitude modulation.

7.8.4 Demodulation of Carrier Digital Signals

- **ASK Detection**

- Just like AM, ASK can be demodulated both coherently (for synchronous detection) or noncoherently (for envelope detection).
- The coherent detector requires more elaborate equipment and has superior performance, especially when the signal power (hence SNR) is low.
- For higher SNR, the envelope detector performs almost as well as the coherent detector. Hence, coherent detection is not often used for ASK because it will defeat its very purpose (the simplicity of detection).

➢ FSK Detection

- Since the binary FSK can be viewed as two interleaved ASK signals with carrier frequencies ω_{c0} and ω_{c1}, *T*herefore, FSK can be detected coherently or non-coherently.
- In non-coherent detection, the incoming signal is applied to a pair of filters tuned to ω_{c0} and ω_{c1}, respectively. Each filter is followed by an envelope detector. The outputs of the two envelope detectors are sampled and compared. If a 0 is transmitted by a pulse of frequency ω_{c0} , then this pulse will appear at the output of the filter tuned to ω_{c0}. Practically no signal appears at the output of the filter tuned to ω_{c1} .

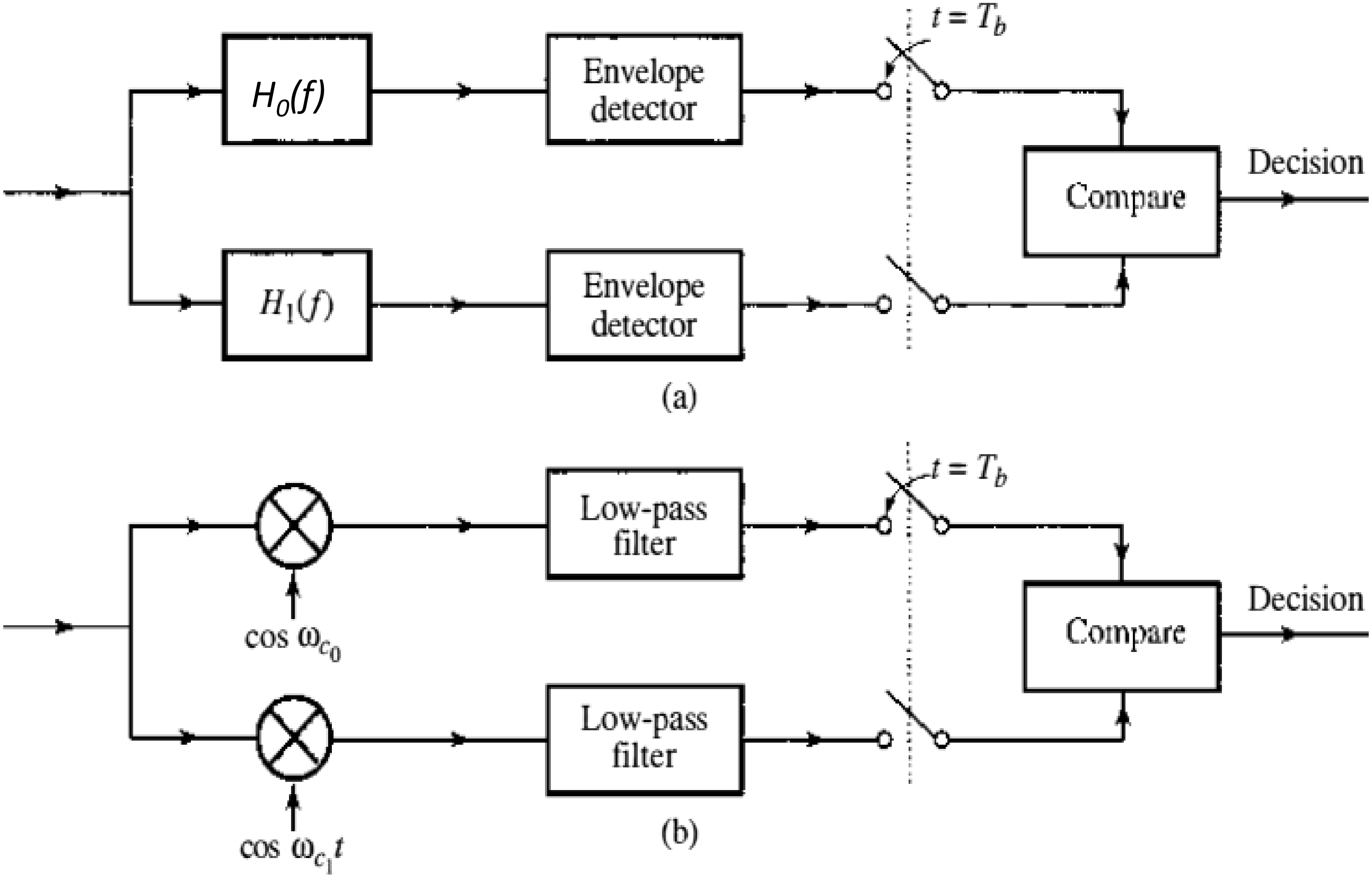


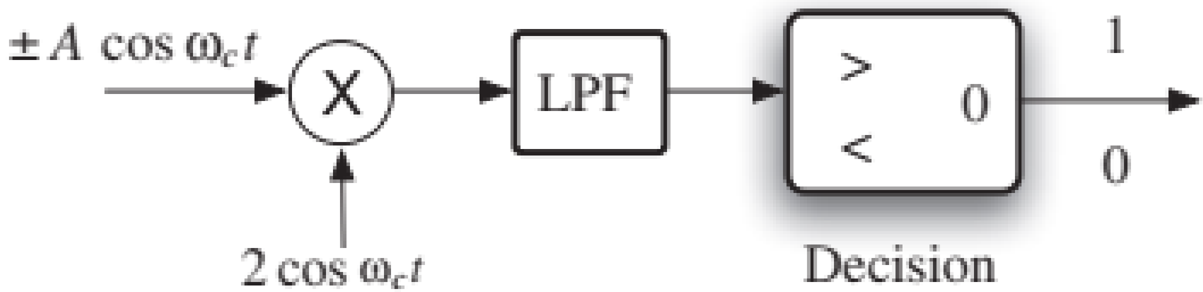
Figure 7.33 (a) Noncoherent detection of FSK.
(b) Coherent detection of FSK

- Hence, the sample of the envelope detector output following the ω_{c0} filter will be greater than the sample of the envelope detector output following the ω_{c1} filter, and the receiver decides that a 0 was transmitted.
- In the case of a 1, the opposite happens.
- FSK can also be detected coherently by generating two references of frequencies ω_{c0} and ω_{c1} , for the two demodulators, in synchronization with the modulation carriers to demodulate the signal received and then comparing the outputs of the two demodulators as shown in Fig. 7.33b. In practice, coherent FSK detection is not in use.

➢ PSK Detection

- Since PSK is a digital manifestation of the DSB-SC amplitude modulation, then PSK must be demodulated coherently only as shown *in Fig. 7.34.*

Figure 7.34 Coherent binary PSK detector (similar to a DSB-SC demodulator).



Differential PSK (DPSK)

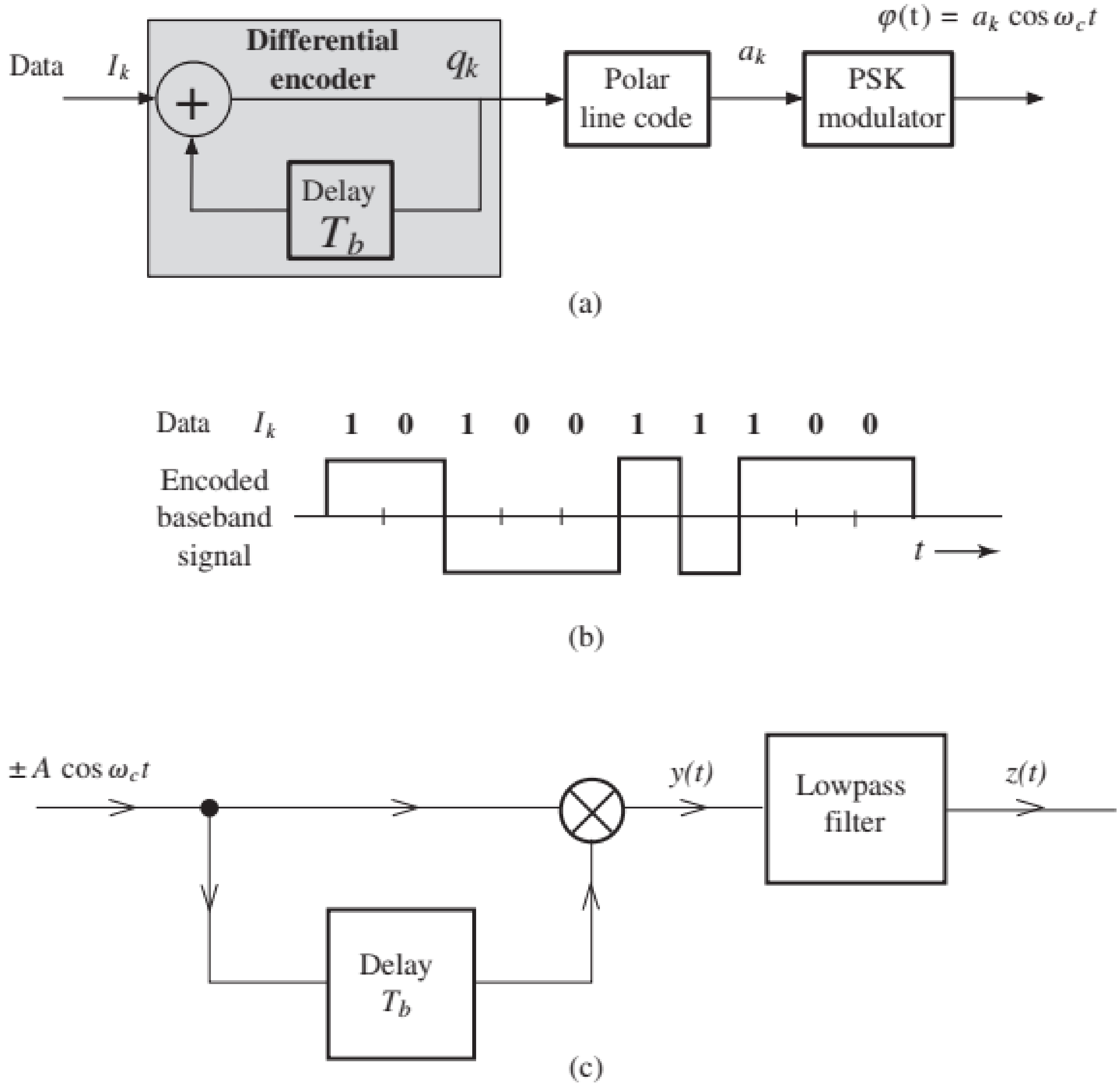
- The main advantage of DPSK is that it can be demodulated noncoherently.
- The principle of differential detection is for the receiver to detect the relative phase change between successive modulated phases θ_k and θ_{k-1}.
- Since the phase value in PSK is finite (equaling to 0 and π in binary PSK), the transmitter can encode the information data into the phase difference $\theta_k - \theta_{k-1}$.
- For example, a phase difference of zero represents 0 whereas a phase difference of π signifies 1.
- DPSK uses differential encoding followed by a polar line coding, and PSK modulation.

- In differential encoding, a 0 is encoded by the same pulse used to encode the previous data bit (no transition), and a 1 is encoded by the negative of the pulse used to encode the previous data bit (transition), as shown in Fig. 7.35a.
- Notice that the addition is modulo-2. The encoded signal is shown in Fig. 7.35b. Thus a transition in the line code pulse sequence indicates **1** and no transition indicates **0.**
- The modulated signal consists of pulses

$$A \cos(\omega_c t + \theta_k) = \pm A \cos \omega_c t$$

- If the data bit is 0, the present pulse and the previous pulse have the same polarity or phase; both pulses are either $A\cos\omega_c t$ or $-A\cos\omega_c t$. If the data bit is 1, the present pulse and the previous pulse are of opposite polarities or phases; if the present pulse is $A\cos\omega_c t$, the previous pulse is $-A\cos\omega_c t$, and vice versa.

***Figure 7.35* (a) Differential encoding; (b) Encoded signal (c) Differential PSK receiver.**



- For demodulation, in place of the carrier, we use the received signal delayed by T_b (one bit interval) as shown in Fig. 7.35c.
- If the received pulse is identical to the previous pulse, the product $y(t) = A^2 \cos^2\omega_c t = (A^2/2)(1 + \cos 2\omega_c t)$, and the low-pass filter output $z(t) = A^2/2$.
  We immediately detect the present bit as **0**.
- If the received pulse and the previous pulse are of opposite polarity, $y(t) = -A^2 \cos^2\omega_c t$ and $z(t) = -A^2/2$, and the present bit is detected as **1**. Table 7.3 illustrates a specific example of the encoding and decoding.
- Thus, in terms of demodulation complexity, ASK, FSK, and DPSK can all be noncoherently detected without a synchronous carrier at the receiver. On the other hand, PSK must be coherently detected.

TABLE 7.3
Differential Encoding and Detection of Binary DPSK

Time k	0	1	2	3	4	5	6	7	8	9	10
I_k		1	0	1	0	0	1	1	1	0	0
q_k	0	1	1	0	0	0	1	0	1	1	1
Line code a_k	−1	1	1	−1	−1	−1	1	−1	1	1	1
θ_k	π	0	0	π	π	π	0	π	0	0	0
$\theta_k - \theta_{k-1}$		π	0	π	0	0	π	π	π	0	0
Detected bits		1	0	1	0	0	1	1	1	0	0

- Noncoherent detection, however, comes with a price in terms of noise immunity. From the point of view of noise immunity, coherent PSK is superior to all other schemes. PSK also requires smaller bandwidth than FSK.

7.9 M-ARY DIGITAL CARRIER MODULATION

- The binary digital carrier modulations of ASK, FSK, and PSK all transmit one bit of information over the interval of T_b second, or at the bit rate of $(1/T_b)$ bit/sec.
- Similar to digital baseband transmission, higher data rate transmission can be achieved either by reducing T_b or by applying M-ary signaling; the first option requires more bandwidth; the second requires more power.
- In most communication systems, bandwidth is strictly limited. Thus, to conserve bandwidth, an effective way to increase transmission data rate is to generalize binary modulation by employing M-ary signaling. Specifically, we can apply M-level ASK, M-frequency FSK, and M-phase **PSK** modulations.

➢ **M-ary ASK and Noncoherent Detection**

- M-ary ASK is a very simple generalization of binary ASK. Instead of sending only

$$\varphi(t) = 0 \text{ for } \mathbf{0} \qquad \text{and} \qquad \varphi(t) = A \cos \omega_c t \text{ for } \mathbf{1}$$

M-ary ASK can send ($\log_2 M$) bits each time by transmitting, for example,

$$\varphi(t) = 0,\ A \cos \omega_c t,\ 2A \cos \omega_c t,\ \ldots,\ (M-1)A \cos \omega_c t$$

- This is still an AM signal that uses M different amplitudes and a modulation index of $\mu = 1$. Its bandwidth remains the same as that of the binary ASK, while its power is increased proportionally with M^2. Its demodulation would again be achieved via envelope detection or coherent detection.

➢ M-ary FSK and Orthogonal Signaling

- M-FSK is similarly generated by selecting one sinusoid from the set $\{A\cos 2\pi f_i t,\ i = 1, \ldots, M\}$ to transmit a particular pattern of $\log_2 M$ bits.
- Generally for FSK, we can design a frequency increment δf and let

$$f_m = f_1 + (m-1)\delta f \qquad m = 1, 2, \ldots, M$$

- For this FSK with equal frequency separation, the frequency deviation (in analyzing the FM signal) is

$$\Delta f = \frac{f_M - f_1}{2} = \frac{1}{2}(M-1)\delta f \qquad \textbf{A}$$

- It is therefore clear that the selection of the frequency set $\{f_i\}$ determines the performance and the bandwidth of the FSK modulation.

- large δf leads to bandwidth waste, whereas small δf is prone to detection error due to transmission noise and interference.
- The minimum δf can be achieved by choosing the FSK signals to be orthogonal in T_s, where $T_s = (\log_2 M) T_b$ is the time interval when different FSK symbols will exhibit virtually no difference such that the receiver will be unable to distinguish the different FSK symbols reliably.
- The orthogonality will guaranties that the FSK signals will be distinct over T_s, and the bandwidth consumption will be small. To find the minimum δf that leads to an orthogonal set of FSK signals, the orthogonality condition according to Sec. 2.7.2 requires that

$$\int_0^{T_s} A\cos(2\pi f_m t) A\cos(2\pi f_n t)\, dt = 0 \qquad m \neq n$$

- We can use this requirement to find the minimum δf. First of all,

$$\int_0^{T_s} A\cos(2\pi f_m t) A\cos(2\pi f_n t)\,dt = \frac{A^2}{2}\int_0^{T_s}\left[\cos 2\pi(f_m+f_n)t + \cos 2\pi(f_m-f_n)t\right]dt$$

$$= \frac{A^2}{2}T_s\frac{\sin 2\pi(f_m+f_n)T_s}{2\pi(f_m+f_n)T_s} + \frac{A^2}{2}T_s\frac{\sin 2\pi(f_m-f_n)T_s}{2\pi(f_m-f_n)T_s}$$

- Since in practical modulations, $(f_m + f_n)T_b$ is very large (often no smaller than 10^3), the first term in Eq. is effectively zero.
- Thus, the orthogonality condition reduces to the requirement that for any integer $m \neq n$,

$$\frac{A^2}{2}\frac{\sin 2\pi(f_m-f_n)T_s}{2\pi(f_m-f_n)} = 0$$

- Because $f_m = f_1 + (m - 1)\delta f$, for mutual orthogonality we have

$$\sin[2\pi(m-n)\delta f T_s] = 0 \qquad m \neq n$$

- From this requirement, it is therefore clear that the smallest δf to satisfy the mutual orthogonality condition is

$$\delta f = \frac{1}{2T_s} = \frac{1}{2(log_2 M)T_b} \qquad \textbf{B}$$

- This choice of minimum frequency separation is known as the minimum shift FSK. Since it forms an orthogonal set of symbols, it is often known as orthogonal FSK signaling.
- Using Equations. (A and B), the frequency deviation Δf for minimum shift FSK is

$$\Delta f = (M-1)/(4T_s)$$

- And its bandwidth according to Carson's rule is approximately

$$2(\Delta f + B) = \frac{M+3}{2T_s} = \frac{M+3}{2T_b \log_2 M} \text{Hz}$$

- The demodulation of M-ary FSK signals follows the same approach as the binary FSK demodulation. Generalizing the binary FSK demodulators of Fig. 7.33 we can apply a bank of *M* coherent or noncoherent detectors to the M-ary FSK signal before making a decision based on the strongest detector branch.
- In fact, it can be in general shown that the bandwidth of an orthogonal *M*-ary scheme is *M* times that of the binary scheme.
- Therefore, in an M-ary orthogonal scheme, the rate of communication increases by a factor of $\log_2 M$ at the cost of M-fold transmission bandwidth increase.
- For a comparable noise immunity, the transmitted power is practically independent of M in the orthogonal scheme. Therefore, unlike M-ary ASK, M-ary FSK does not require more transmission power. However, its bandwidth requirement increases almost linearly with M (compared with binary FSK or M-ary ASK).

- The minimum shift FSK can be described geometrically by applying the concept of orthonormal basis functions in Sec. 2.6.

$$\psi_i(t) = \sqrt{\frac{2}{T_s}} \cos 2\pi \left(f_1 + \frac{i-1}{2T_s} \right) t$$

for i = 1, 2,…, M.

- It can be simply verified that

$$\int_0^{T_s} \psi_m(t)\psi_n(t)\,dt = \begin{cases} 1 & m = n \\ 0 & m \neq n \end{cases}$$

Thus, each of the FSK symbols can be written as

$$A\cos 2\pi f_m t = A\sqrt{\frac{T_s}{2}}\psi_m(t) \qquad m = 1, 2, \ldots, M$$

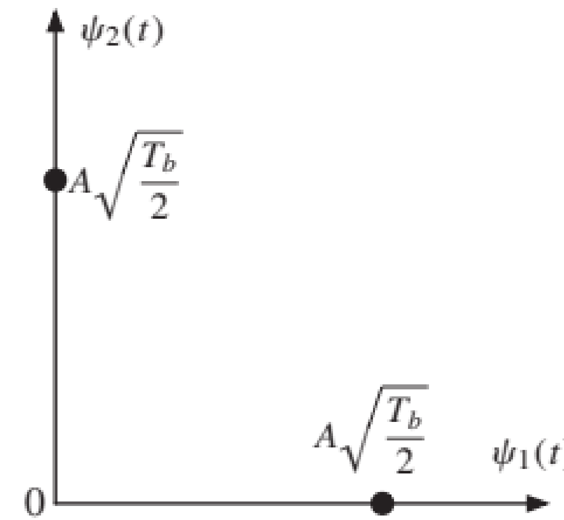


Figure 7.36 Binary FSK symbols in the 2D orthogonal signal space.

- The geometrical relationship of the two FSK symbols for M = 2 is easily captured by Fig. 7.36.